MILITARY SPECIFICATION

WHEEL AND CARBON BRAKE ASSEMBLY,

F-14A

GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for construction, performance and test of the Wheel and Brake Assembly of the F-14A Weapon System. Specification MIL-W-5013 shall apply except as modified herein.

1.2 Applicable Drawing. The wheel and brake assembly shall conform to Fig. 2.

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-S-5002	Surface Treatments and Metallic Coatings for Metallic Surfaces of Weapons Systems
MIL-W-5013	Wheel and Brake Assemblies; Aircraft
MIL-H-5440	Hydraulic Systems, Aircraft Types I and II, Design Installation and Data Requirements for
MIL-C-5503	Cylinders: Aeronautical, Hydraulic Actuating General Requirements for

Beneficial Comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Air Systems Command, Washington, D.C. 20361, Attention: AIR-530321 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1630

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MIL-P-5514	Packings, Installation and Gland Design, Hydraulic, General Requirements for
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-H-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile and Ordnance
MIL-C-6021	Castings, Classification and Inspection of
MIL-B-6812	Bolts, Aircraft
MIL-I-6866	Inspection, Penetrant, Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-6875	Heat Treatment of Steels Aerospace Practice, Process for
MIL-F-7179	Finishes and Coatings: Protection of Aerospace Weapons Systems, Structures and Parts
MIL-B-7838	Bolt, Internal Wrenching 160 KSI FTU
MIL-B-7874	Bolt, Machine, 1200°F
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-8699	Steel Bars and Forging Stock (4330) Vanadium Modified, Aircraft Qualified
MIL-H-8775	Hydraulic System Components, Aircraft and Missiles General Specifications for
MIL-B-8831	Bolt, 12 Point, External Wrenching 180,000 PSI
MIL-C-8837	Coating, Cadmium (Vacuum Deposited)
MIL-S-8844	Steel Bars, Reforging Stock and Mechanical tubing, Low Alloy, High Quality

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MIL-S-8879	Screw Threads, Controlled Radius, Root with Increased Minor Diameter General Specification for
MIL-L-8937	Lubricant, Solid Film, Heat Cured
MIL-S-8949	Steel Bars, Plates, Sheets, Billets, and Reforging Stock Type D6AC
MIL-I-8950	Inspection, Ultrasonic, Wrought Metals Processes for
MIL-T-9047	Titanium and Titanium Alloy Bars and Forging Stock
MIL-T-10727	Tin Plating, Electrodeposited or Hot-Dipped for Ferrous and Non-Ferrous Metals
MIL-F-18264	Finishes: Organic Weapons Systems Application and Control of
MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format For
MIL-C-21180	Aluminum Alloy Castings, High Strength
MIL-C-22751	Coating System, Epoxy- Polyamide, Chemical and Solvent Resistant
MIL-A-22771	Aluminum Alloy Forgings. Heat Treated
MIL-P-23377	Primer Coating: Epoxy - Polyamide, Chemical and Solvent Resistant
MIL-B-23964	Bolt, Self-Retaining, Positive Locking
MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range
MIL-L-81352	Lacquer: Acrylic (For Naval Weapon Systems)

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MIL-C-81706	Chemical Conversion Material for Coating Aluminum and Aluminum Alloys
MIL-F-81907	Coating System, Polyurethane, Aliphatic Weather Resistant, Process for Application
MIL-P-81958	Pressure Relief Devices, Overpressurization Prevention, Wheel/Tire Aircraft
MIL-P-81985	Peening of Metals
MIL-H-83282	Fire Resistant Hydraulic Oil

STANDARDS

Military

MIL-STD-130	Identification Marking of Military Property
MIL-STD-143	Specification and Standard Order of Precedence for the Selection of
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing of
MIL-STD-882	System Safety Engineering of Systems and Associated Subsystems and Equipment, General Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipments and Facilities
MS14152	Tire - Aircraft 37 X 11.5 - 16 Type VII (Navy)
MS14155	Washer, Recessed Countersunk Two Surfaces, for use with wheel bolts and nuts up to and including 180 KSI Ftu
MS14156	Nut, Wheel, Self-Locking, Flanged, Steel, 180 KSI Ftu, 450°F Spline Drive

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MS14157	Bolt, Wheel, Tension Flanged Steel, 180 KSI Ftu, 450°F External Spline Drive
MS14163	Bolt, Wheel, Tension, Flanged, Steel 220 KSI Ftu, 450°F External Wrenching Spline Drive
MS14164	Nut, Wheel, Self-Locking Flanged, Steel, 220 KSI Ftu, 450°F, Spline Drive
MS15795	Washers, Flat-Metal, Round, General Purpose
MS20002	Washer, Countersunk and Plain, High Strength
MS28774	Retainer, Packing, Backup, Single-turn, Tetrafluoroethylene
MS28778	Packing, Preformed, Straight Thread Tube Fitting Boss
MS33649	Boss, Fluid Connection, Internal Straight Thread
Air Force-Navy Aeronautical	
AN960	Washer
AND10573	Rim Contours - 26 X 6.6, 30 X 7.7, 32 X 8.8, 34 X 9.9, 36 X 11, 38 X 11, 40 X 12, 42 X 12, 44 X 13 and 46 X 14, Type VII Extra High Pressure, Main Landing Gear Wheels

PUBLICATIONS

Military

ANA438	Age Controls of Age Sensitive Items
WR-62	Naval Weapon Requirements Specifications and Standards, Use of

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(When requesting military specifications, standards, drawings and publications, refer to both title and number. Copies of applicable specifications required in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

Commercial

ASAB46.1	Surface Texture
AMS6419	Steel Bars, Forgings and tubing 1.6 Si-0.82Cr 1.8Ni-0.4Mo -0.07V (0.40 -0.45c) Premium Quality Consumable Electrode Melted

(Copies of listed commercial documents may be obtained from United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Preproduction. - This specification makes provision for preproduction testing. Specification MIL-W-5013 shall apply except as modified herein.

3.2 Selection of Materials, Parts and Processes. - In the selection of materials, parts and processes, fulfillment of major design objectives shall be of the prime consideration. Specifications and standards for all materials, parts and processes which are not specifically designated herein shall be selected in accordance with Standard MIL-STD-143 and Military Publication WR-62 and shall require procuring activity approval prior to use. Consideration shall be given in the selection of materials and processes to the prevention of premature failures due to detrimental effects such as stress corrosion, corrosion, embrittlement and carbide precipitation. Selection of materials shall be made with special attention given to the stringent requirements of temperature, temperature gradients, thermal conductivity, vibration, loads and other life cycle environmental conditions plus the factors of vulnerability, reliability, repairability and lightweight construction.

3.2.1 Materials.- Unless otherwise approved by the procuring activity, material selection shall be made in accordance with paragraph 3.2 of MIL-W-5013. All parts of the assembly shall be made of corrosion resisting material or shall be suitably protected against corrosion internally and externally.

3.2.1.1 Dissimilar Metals. - The use of dissimilar metals in intimate

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contact with each other shall be avoided unless suitably protected against galvanic corrosion. Dissimilar metals are defined in Standard MIL-STD-889. MIL-F-7179 para. 3.1.1.1 shall apply. Carbon composites shall be considered as graphite for dissimilar metal purposes.

3.2.1.2 Steels. -

3.2.1.2.1 Selection of Steels.- Free machining carbon or stainless steel shall not be used. Selection of steels shall be as follows:

(a) Aircraft quality, consumable electrode vacuum melted steel shall be used for all parts made from heat treated alloy steel. Where low alloy steels are used, 4330 V (Mod) in accordance with Specifications MIL-S-8699 and D6AC in accordance with Specification MIL-S-8949 shall be used in the 220,000 through 240,000 psi ultimate strength range. Type 4340 in accordance with Specification MIL-S-8844 shall be used in the 260,000 through 280,000 psi ultimate strength range. Type 300 M in accordance with Specification AMS6419 shall be used in the 280,000 through 300,000 psi ultimate strength range. The ultimate strength range variation for parts over 220,000 psi shall be -0 + 20,000 psi. Parts which are inaccessible, i.e., not readily inspectable or replaceable shall not be heat treated over 220,000 to 240,000 psi ultimate strength range. All low alloy steel shall be heat treated in accordance with Specification MIL-H-6875. All high strength steel fittings heat treated to 220,000 psi and above shall be saturation shot peened in accordance with Specification MIL-P-81985. The use of steel heat treated in excess of 220,000 psi, however, shall be subject to specific approval of the procuring activity.

(b) Preference shall be given, in selection of carbon and low alloy steels, to compositions having the least hardenability which will ensure through hardening of the part concerned.

(c) Compositions shall be selected such that heat treatment to the required strength and service temperatures shall preclude temper-embrittlement. Tempering temperature for selected steels shall be at least 100°F above the maximum service temperature.

(d) Steels shall be selected having ductile-brittle fracture transition temperatures as determined by impact test below the minimum operating temperature.

(e) Steels whose mechanical properties are developed by cold deformation shall be so selected that the recovery temperature will be at least 50°F above the expected operating temperature range.

(f) In the event that production processing and facilities do not permit the complete elimination of surface decarburization on

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surfaces not machined after heat treatment, the design fatigue strength shall be correspondingly reduced and the design procedure shall be as approved by the procuring activity. If any process is employed for a removal of decarburization other than by mechanical means, a process specification shall be submitted to the procuring activity for approval.

(g) The mechanical drilling of holes in martensitic steels after hardening to strength levels of 180,000 psi and above shall be avoided whenever practicable. When drilling is performed on 4330 V (Mod), 4340 and steel parts heat treated to 180,000 psi and above or on D6AC steel parts heat treated to 180,000 psi and above, the final hole sizing shall be performed in accordance with procedure approved by procuring activity. Warpage during heat treatment shall be minimized. Any necessary straightening of parts after heat treatment to strength levels of 180,000 psi and above shall be accomplished in accordance with Specification SD24K, para. 3.2.2.1.2.1(8).

(h) Drilling and machining operations after final heat treatment shall be held to a minimum since burning, formation of untempered martensite and other metallurgical damage may occur. When such operations are necessary, detail processing and inspection procedures require procuring activity approval prior to use. Inspection procedures for parts heat treated to 260,000 psi and above shall include provisions for both fluorescent magnetic particle inspection in accordance with Specification MIL-I-6868 and shall be nital surface etched. Any evidence of burning, formulation of untempered martinsite or metallurgical damage shall be cause for rejection.

(i) Grinding shall be accomplished whenever possible prior to the final tempering operation. When parts are ground subsequent to any tempering operation detail processing and inspection procedures require procuring activity approval prior to use. Inspection procedures for parts heat treated to 220,000 psi and above shall include provisions for both fluorescent magnetic particle inspection in accordance with Specification MIL-I-6868 and shall be nital surface etched. Any evidence of burning, formation of untempered martensite or metallurgical damage shall be cause for rejection.

(j) Ultrasonic inspection requirements for steel products shall be in accordance with Specification MIL-I-8950, Class AAA.

3.2.1.2.2 Corrosion Resistant Steels.- The following limitations shall apply in the selection and application of corrosion resistant steels:

(a) Unstabilized austentic steels shall not be fusion welded.

(b) Cold rolled austenitic stainless steels in work hardened tempers (1/4 hard, etc.) shall not be used at temperatures higher than 50°F below the stress relieving temperatures.

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(c) The following restrictions apply to the use of precipitation hardened stainless steels:

(1) PH steels shall be limited to alloys and alloy tempers known to possess a high resistance to stress corrosion cracking.

(2) AM 355 all tempers, AM 350 SCT 850, 17-4PH below H1025, 15-5 PH H900, 17-7PH RH950 shall not be used.

(3) 17-4PH and 15-5PH shall not be used in areas where high sustained tension could occur when heat treated above 155,000 psi ultimate strength level (RC 35-41). The hardening temperature shall not be lower than 1025°F for less than four hours (condition H1025).

(4) PH steels shall not be used over 600°F or at a strength level of 170,000 psi or over ultimate tensile strength without specific approval.

(5) 17-7 CH900 is acceptable for spring applications.

(d) Types 416, 431 or 19-9DL stainless steel shall not be used.

3.2.1.3 Aluminum. - The following requirements shall be observed.

(a) High strength aluminum forgings shall be in accordance with MIL-A-22771.

(b) If ultrasonic inspection is performed either on the original forging billet, or at an intermediate forming stage, the final machined forgings need not be. However, 100 percent penetrant inspection in accordance Specification MIL-I-6866 shall be performed on fully machined parts.

(c) All high strength aluminum fittings shall be saturation shot peened in accordance with Specification MIL-P-81985.

(d) When aluminum sheet metal parts are required, spot or seam welding of aluminum shall not be used.

(e) 2000 series aluminum alloys in the T3X-T4X tempers shall not be used. 7000 series aluminum alloys in the T6 temper shall not be used.

(f) Where aluminum castings are required, they shall be in accordance with MIL-A-21180 unless otherwise approved by procuring activity.

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3.2.1.4 Titanium.- The following requirements shall apply:

(a) Titanium and titanium base alloys may be used in applications where their use is justified in terms of weight savings, improved performance, improved serviceability and where adequacy of manufacturing methods can be demonstrated. All applications shall use the annealed rather than the solution treated or solution treated and aged material condition. The use of titanium or titanium base alloys require specific procuring activity approval prior to use. All titanium machined parts shall be saturation shot peened in accordance with Specification MIL-P-81985. This requirement may be waived for non-fatigue susceptible parts upon which no grinding operation is performed, subject to procuring activity approval.

(b) Titanium alloys, 6Al-4V and 6Al-6V-2Sn are preferred. Procurement of forging of these alloys shall be in accordance with Specification MIL-T-9047. In areas where welding is desired, Ti 6Al-4V shall be used. All welds shall be made by electron beam techniques and shall be machined flush on both sides except with specific approval of the procuring activity. Resistance spot welding may be used on titanium provided substantiating data is submitted and specific approval is obtained from the procuring activity.

(c) The use of titanium alloys Ti 8Al-1Mo-1V and Ti 13V-11Cr-3Al is prohibited.

(d) The use of solvents or cutting fluids which are chlorinated or sulfonated is prohibited.

(e) Grinding of machined parts shall only be used if no alternate method of metal removal is available. Grinding or sanding is not permitted on surfaces that are not subsequently shot peened. Grinding of sheet is acceptable provided a minimum of .003 inch is chemically milled from the ground surface.

(f) Ultrasonic inspection requirements for titanium products shall be in accordance with Specification MIL-I-8950, Class AAA.

3.2.1.5 Magnesium. - Magnesium and magnesium alloys shall not be used.

3.2.1.6 Beryllium. - Shall not be used.

3.2.1.7 Carbon Composites. - Structural carbon composites shall be used for heat sink material. The carbon heat sink material shall be traceable by batch or lot number to the brake serial number level. The Seller shall develop material consistency tests for NAVAIR approval which shall be conducted on representative material samples per 4.4.2. The Seller

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established material properties tests will be conducted on a batch sample of the qualification test hardware in order to define the baseline properties for quality control of follow-on brake heat sinks.

3.2.1.8 Castings. The use of castings for structural applications requires procuring activity approval prior to use.

3.2.1.9 Fittings. - Structural fittings shall be made from aluminum, steel or titanium alloys within the limitations imposed per this specification or other applicable specifications. Connections of solid end-fittings to wheel and brake assembly using aluminum alloy rivets shall be suitable for possible replacement of rivets by the next larger size. Steel rivets or bolts may be substituted for aluminum alloy rivets subject to approval by the procuring activity. Abrupt changes in cross section shall be avoided. The minimum fillet radius for structural parts shall be 0.110 inch. Where justified by design, if not critical in fatigue, smaller radii may be used subject to the procuring activity approval. This requirement also applies to spot faces, counterbores, countersinks, and recesses.

3.2.1.9.1 Threads for Fittings. - In case of structural fittings produced of steel, heat-treated in excess of 125,000 psi and incorporating a threaded portion loaded primarily in tension, the threads shall be in accordance with Specification MIL-S-8879. The threads shall be rolled in a single pass after heat treatment. Where the state-of-the-art does not permit this, specific approval is required.

3.2.2 Parts.

3.2.2.1 Standard Parts.- Standard parts (MS,AN, or JAN) shall be used whenever they are suitable for the purpose and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts and cotter pins shall not be used without specific approval of the procuring activity.

3.2.2.2 Bushings. - Bushings shall be provided for all bolts or pins subject to angular or other motions which would tend to distort or enlarge the hole. Bushings shall be securely anchored (an interference fit is acceptable) to the member to preclude slippage or movement. Bushings shall assume all wear or deformation at the joint and be readily replaceable. Peening or staking is prohibited. A bushing, however, with a very close sliding fit may be used as a sliding spacer to take up accumulated width tolerances so a fitting will not be deformed due to torquing with attachment bolt, for example: When using a sliding bushing to clamp the inner race of a bearing without deforming the basic fitting. In the event the inside diameter of a bushing is distorted out of round during the press fit operation, the bushing shall be reamed to size after installation.

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Reestablishment of the finish after reaming is required. Where the shoulders of two bushings are in sliding contact, the shoulder of one of the bushings shall be hard chrome plated or otherwise treated to form a suitable bearing surface.

3.2.2.3 Bearings. - Bearings shall be in accordance with Specification MIL-W-5013.

3.2.2.4 Bolts. - Bolts heat treated to a minimum of 125,000 psi for general structural applications shall be in accordance with Specification MIL-B-6812. Bolts heat treated from 160,000 psi to 180,000 psi shall be in accordance with Specification MIL-B-7838. Bolts heat treated from 180,000 to 200,000 psi shall be in accordance with Specification MIL-B-8831. High strength bolts of greater than 200,000 psi ultimate tensile strength may be used subject to the procuring activity approval. Approved type high strength (160,000 psi minimum ultimate tensile strength) internal wrenching bolts shall be in accordance with Specification MIL-B-7838. Corrosion resisting steel bolts in temperatures not exceeding 1200°F, shall be in accordance with Specification MIL-B-7874, as applicable. Steel bolts smaller than one-fourth inch in dismeter shall not be used in any single bolted structural connection or any application where a failure would adversely affect safety of flight. Aluminum alloy bolts, nuts, and screws may be used in nonstructural lightly stressed aluminum alloy parts. Structural bolts which are loaded in tension shall be pre-stressed to a value consistent with minimizing the effects of fatigue in the joint. The proper bolt torque values shall appear on the applicable installation drawings. Where it is necessary to use a single attachment bolt with the head down in an application where its loss would affect safety of flight, the head of the bolt shall be lockwired or retained in position independent of the attaching nut. Cadmium plated steel bolts or nuts used with aluminum alloy parts shall be insulated from the aluminum alloy by aluminum alloy washers beneath the bolt head and nut, except that cadmium-plated steel washers may be used for bolts loaded in tension. Titanium bolts may be used subject to the following design and usage limitations:

(a) Titanium fasteners shall not be used where they are subject to primary tension loading.

(b) Titanium fasteners shall not be used for single joint connections such as control rod end attachments or structural pin joints unless the fasteners have been coated with an approved dry film lubricant (MIL-L-8937, etc.).

(c) Designs requiring motion on the surface of titanium bolts (e.g. rotating joints) shall be avoided. Titanium sliding against itself or any other material such as stainless steel or alloy steels will gall readily. This creates stress concentrations which lead to early fastener failure or seizing of the joint.

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(d) Titanium bolts may be used only in conjunction with steel nuts that have an approved type dry film lubricant. However, Hi-Lok type fasteners may be used with either aluminum or steel collars.

(e) Titanium bolts smaller than one-fourth inch in diameter shall not be used in any application where a failure could occur as defined in 3.3.11 herein.

(f) When titanium fasteners are required and the choice has to be made between Hi-Loks and Lock-bolts, Hi-Loks shall be used.

(g) Titanium nuts shall not be used.

3.2.2.4.1 Close-Tolerance Bolts. - Close-tolerance series bolts or pins shall be used in shear connections of the wheel and brake assembly except where the number of bolts is sufficient to obtain a group effect and the connection is such as not to require the improved fatigue characteristics of close tolerance fasteners.

3.2.2.4.2 Safety Bolts. - Bolted joints shall be safetied. Bolts shall be of proper grip and length in order to prevent looseness or binding.

3.2.2.4.3 Self-Retaining Bolts. - Consideration shall be given to the use of self-retaining bolts in any joint which will require frequent disassembly for maintenance and is either of single attachment, serves as an axis of rotation, or is designed to transmit motion which may result in relative rotation between the components of the joint. The bolts shall be additionally locked in position by self-locking castellated nuts. Self-retaining bolts shall be in accordance with Specification MIL-B-23964.

3.2.2.5 Pins and Fasteners.

(a) The use of friction-retained pins without auxiliary means of retention is prohibited (e.g., groove pins, taper pins and etc.). Peening, staking or safety wiring is not acceptable for pin retention. Roll pins are absolutely prohibited.

(b) Rotating pins, or bolts, shall be hard chrome plated in accordance with procedures noted herein.

(c) Approved titanium and titanium base alloy fasteners may be used subject to restrictions of 3.2.2.4.

3.2.2.6 Washers.- Washers used with internal wrenching or other similar high strength tensile type bolts shall be in accordance with

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Standard MS20002, except in those cases where special washers, chamfered on both sides, are required to prevent improper installation. Washers used with other structural fasteners shall be in accordance with Standard AN960 or MS15795. Lock washers shall not be used. Metallic crush washers shall not be used for sealing purposes or any other application.

3.2.3 Stress Corrosion Factors. - To prevent premature failures caused by stress corrosion or hydrogen embrittlement, the design and method of manufacture of parts of steels heat treated to tensile strengths above 220,000 psi, or of bare high strength aluminum alloys or titanium alloys and the techniques by which they are assembled and installed shall be such that sustained or residual surface tensile stress and stress concentrations are minimized. Practices, such as: The use of press or shrink fits; taper pins; clevis joints in which tightening of the bolt imposes a bending load on the lugs; and straightening and assembly operations; which result in sustained or residual surface tensile stresses shall be avoided wherever practicable. In cases where such practices cannot be avoided, corrective practices such as stress relief heat treatments, optimum grain flow orientation, shot peening, or similar surface working shall be used to minimize the hazard of stress corrosion or hydrogen embrittlement damage.

3.2.4 Fatigue Factors.- To prevent premature failures caused by repeated loads, the design and method of manufacture, including consideration of the damaging effect of decarburization and certain metallic coatings of all critical parts and the techniques by which they are assembled and installed shall be such that sustained or residual tensile stresses and stress concentration are minimized. Practices such as cold straightening, cold forming, and the assembly of mismatched surfaces, which result in sustained or residual surface tensile stresses shall be avoided. In cases where such practices cannot be avoided, corrective practices such as stress relief heat treatment, optimum grain flow orientation, shimming, shot peening or similar surface working shall be used to minimize premature fatigue failure, subject to approval by the procuring activity. Surface roughness of all surfaces which are subject to repeated stresses, shall not be in excess of 125 rms as defined in Standard ASAB46.1. Particular attention shall be given to optimum heat treatment procedures, corrosion protection and finish to minimize corrosion damage which may be the site of premature fatigue failure.

3.2.5 Finish. - The wheel and brake assembly and its components shall be finished in accordance with the following.

3.2.5.1 Surface Treatments.

3.2.5.1.1 Aluminum. - All aluminum and aluminum alloy parts shall be completely anodized in accordance with Specifications MIL-A-8625,

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Type II except where hard coat may be necessary or beneficial to proper function of brake assembly, such as pistons, to reduce galling if seal is contained in the cylinder wall. Hardcoat shall be in accordance with Specifications MIL-A-8625. For touch up of anodized surfaces which were marred during minor rework, assembly or installation, a chemical film may be used in accordance with Specifications MIL-C-5541.

3.2.5.1.2 Steel. - Steel shall be protected as follows:

(a) Chromium Plating - In conformance with Specifications QQ-C-320, 240,000 psi and over, bake at 375 ± 25°F within three hours after plating for a minimum of 23 hours. After grinding of chrome plate, 220,000 psi and over, all parts shall be baked at 375 ± 25°F for three hours. All ground chrome plated parts shall be fluorescent magnetic particle inspected in accordance with Specification MIL-I-6868.

(b) Parts heat treated to 220,000 - 240,000 psi range shall have at least an eight hours bake for embrittlement relief in accordance with Specifications QQ-C-320.

(c) The minimum thickness shall be 0.002 inches for all chrome plated parts except the piston, which shall be 0.0035 inches.

(d) Prior to plating, critical parts such as pins, pistons and cylinders shall be shot peened.

(e) Plating shall be applied directly on steel and at a rate not to exceed 0.0005 inches per hour. All external non-corrosion resisting steel parts except as noted, shall be cadmium plated in accordance with Specification QQ-P-416, Type II, Class 2, except as follows:

(f) Steel parts heat treated to 220,000 - 240,000 psi range, which are to be cadmium plated, shall be vacuum coated in accordance with Specification MIL-C-8837. Any other process shall be subject to the procuring activity approval.

(g) Tin Plating. - Tin plating, in accordance with Specifications MIL-T-10727, Type I, shall be used in lieu of cadmium plating where flaking may tend to contaminate equipment in contact with hydraulic fluid.

3.2.5.1.2.1 Corrosion Resistant Steels. - Corrosion resistant steels shall be passivated in accordance with Specification MIL-S-5002.

3.2.5.1.3 Titanium. - Surface treatment of titanium and titanium alloys shall be as approved by the procuring activity prior to use.

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3.2.5.1.4 Miscellaneous. - Surface treatments other than those noted shall be subject to the procuring activity approval prior to use.

3.2.5.2 Priming and Painting. - Priming and painting shall be in accordance with MIL-F-7179.

3.2.5.2.1 Plates. - The backs of mechanically attached name plates, instruction plates, and designation plates shall be primed in accordance with Specification MIL-F-23377 epoxy polyamide primer prior to installation. Upon installation, the rear of these plates shall be sealed in accordance with Specification MIL-S-8802, Class B sealant. The faces of all plates shall be covered with a coating of clear acrylic lacquer conforming to Specification MIL-L-81352 and applied in accordance with Specification MIL-F-18264.

3.2.5.3 Non-Specification Material - Materials for which no Federal, military, or industry specification exists, the contractor shall be required to develop specifications covering technical requirements and test methods thereof and acceptance criteria for review and acceptance by procuring activity.

3.2.6 Packing, "O" Rings and Gaskets. - Packings, "O" rings and gaskets shall be in accordance with the following modifications:

Wheels

(a) "O" rings and standard packing glands shall be in accordance with Specification MIL-P-5514.

(b) Finish on seal surfaces shall be at least as fine as that specified in Specification MIL-C-5503.

Brakes

(a) "O" rings shall be in accordance with Specification MIL-P-5514 with the following exceptions:

Operating temperatures of "O" rings shall be 300°F for continuous use. Short term (1 hour) exposure to 450°F shall not result in leakage at maximum operating brake pressure.

(b) High temperature seals may be used but must be installed in a standard groove.

(c) Back-up rings if used shall conform to MS 28775. Teflon compounds containing up to 15% graphite may be used for back-up rings.

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3.2.7 Foolproofness. - The wheel and brake assembly shall be as foolproof as possible to avoid incorrect assembly which would result in damage or malfunction. Joints shall be so arranged that incorrect assembly will be extremely difficult, if not impossible. Brackets, levers, links, bearings, and control rods that can be installed backwards or upside down shall, whenever practicable, be symmetrical about each of the three (3) axes, or it shall be impossible to install unsymmetrical parts upside down or backwards. All electrical, fluid, or mechanical connections for controls and hydraulic lines, shall be incapable of being reinstalled incorrectly where such incorrect reinstallation would involve damage or malfunctioning.

3.2.8 Interchangeability. - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable as defined in Specification MIL-I-8500.

3.3 Design and Construction. - The wheel and brake assembly shall be designed and constructed for installation in accordance with the envelope and mounting provisions of Figure 2.

3.3.1 Installation. - The wheel and brake assembly shall be capable of installation on the left and right hand sides of the aircraft, both as a wheel and brake assembly and as separate wheel assemblies and brake assemblies.

3.3.1.1 Hub Cap. - Provisions for driving an anti-skid sensor on the hub cap shall be provided in accordance with Figure 3. The hub cap shall be furnished as part of the wheel assembly.

3.3.2 Anti-Skid. - The wheel and brake assembly shall be designed to operate satisfactorily with an anti-skid control system. Provisions for mounting the wheel speed sensor unit shall be supplied within the axle.

3.3.3 Contaminant Protection. - Consideration shall be given during design to prevent entry of excessive amounts of mud, dust, and dirt into vulnerable areas of the brake assembly.

3.3.4 Wheel and Construction.

3.3.4.1 Wheel Assembly. - The wheel assembly shall be the divided type secured by the bolts and self-locking nuts. Tie bolts and nuts shall be selected with consideration to: best fatigue life, equal strength bolt and nut, light weight, and serviceability. Double chamfered washers of the same design are required under the tie bolt heads and the bolt nuts.

3.3.4.2 False Axle. - The wheel assembly shall provide an integral false axle and false axle nut positively secured to prevent unlocking.

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3.3.4.3 Heat Shield. - A heat shield, if required, shall be provided to minimize direct radiation between the brake heat sink and the wheel. In addition, the heat shield shall minimize conducted and convected heat flow. The net effect shall be to limit the temperature of the tire to a level which will not cause tire deterioration, blowouts, or excessive melting of the thermal relief plugs. The heat shield shall be designed to minimize possible damage due to normal wheel, tire, and brake maintenance and handling.

3.3.4.4 Tire. - The wheel assembly shall be designed for use with a 37 X 11.50-16 tire conforming to drawing MS 14152.

3.3.4.5 Wheel Fuse Plugs. - A minimum of 3 fuse plugs (thermal sensitive pressure release devices) shall be provided in the wheel at the areas of most consistent temperature rises. The devices shall be so located that their replacement requires tire removal. Fuse plugs shall be sized to preclude tire blowout during, or after, the requirements in Table I, Condition A and B and 4.3.1.2.8.3. The fuse plug temperature setting shall be subject to NAVAIR concurrance.

3.3.4.6 Tie Bolt Bosses. - Bosses shall be raised, where possible.

3.3.4.7 Lubrication. - The wheel bearings shall be lubricated with grease conforming to Specification MIL-G-81322.

3.3.4.8 Wheel Rim. - The wheel rim contours shall be in accordance with Standard AND 10573.

3.3.4.9 Wheel Bolts.- The wheel tie bolts, nuts and washers combination shall be as follows:

(a) Tie bolt - MS 14157 or MS 14163
(b) Nut - MS 14156 or MS 14164
(c) Washer - MS 14155

3.3.4.10 Tire Inflation Valve.- The inflation valve shall be the over pressurization relief type conforming to MIL-P-81958 (AS), Type IV, to prevent inadvertent overpressurization of the tires during servicing.

3.3.4.11 Wheel/Brake Clearance. - The vendor shall submit a report to NAVAIR for approval showing adequate radial clearance for all load conditions in this specification via finite element deflection analysis of the wheel, brake and axle.

3.3.4.12 Experimental Stress Analysis. - Experimental stress analysis will concentrate effort toward a minimum weight wheel. A photo

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stress survey or other means shall be performed during the design phase to locate areas of critical stress and verify the wheel design. A stress analysis report of the wheel shall be submitted to NAVAIR for approval.

3.3.5 Brake Construction.

3.3.5.1 The brake shall employ a structural carbon heat sink. Rotor lugs must use clips or metallic structure to protect against damage during wheel installation. Chamfered entry shall be used on wheel keys and/or clips to facilitate alignment during wheel installation. Exposed surfaces of the carbon heat sink shall be adequately protected from oxidation.

3.3.5.2 Piston Housing. - The brake piston housing shall contain an integral, non-interflow shuttle valve. The internal spool shall be free to float. Performance of the valve shall conform to the requirements of MIL-V-19068A and the following:

(a) The shuttle valve shall be designed to prevent hang up or stopping in an intermediate position through all operating pressure-temperature ranges.

(b) Operating Range

(1) Intermittent, Non-operating 450°F for 4 minutes and 300°F for 20 minutes.

(2) Continuous Operating -65°F to +160°F

(3) Operating Pressure 3000 psi; Proof 4500 psi; Burst 6000 psi (Min)

3.3.5.3 Hydraulic System. - The brake shall be designed for use with hydraulic fluids, Specifications MIL-H-5606 and MIL-H-83282, in a 3000 psi, Type II system per Specification MIL-H-5440 providing five micron absolute filtration.

3.3.5.4 Fluid Temperature. - The hydraulic fluid temperature in the brake piston cavities shall not exceed 300°F after brake stops during which the kinetic energy absorbed is equal to or less than that for Test Condition A or B, Table I.

3.3.5.5 Pistons. - The pistons shall have their own inlet port and bleeder fitting. Inlet ports shall conform to MS33649 and bleeder fittings shall be made of steel, tin-plated for Specification MIL-T-10727, Type I or cadmium plated in accordance with Specification MIL-C-8837 subject to procuring activity approval. The brake shall properly bleed when mounted with its symmetrical centerline within 5° of

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vertical. Inlet ports, bleeder fittings and hydraulic lines shall be so located as to preclude damage from arresting cables. Piston stops shall be provided which limit piston travel to prevent venting of hydraulic fluid upon stack failure or operation beyond the normal removal condition. The piston stops shall allow sufficient piston travel to permit a MDGW RTO at the worn out brake condition (flush wear pin) at 3000 psi of hydraulic pressure applied, or .100 inch travel beyond the wear out limit (flush wear pin) whichever is greater. The stops shall be designed for 4500 psi piston pressure without the disks installed.

3.3.5.6 Running Clearance. - The built in running clearance shall be sufficient to comply with paragraph 3.3.5.10 for new and used brakes and for any hot and cold brake conditions or pressurization histories through Condition B of Table 1.

3.3.5.7 Displacement. - The maximum fluid displacement of the brake shall not exceed 2.5 cubic inches at 3000 psi. The displacement after brake contact shall be minimized to optimize pressure control, at the brake, with minimum flow. There shall be no increase in hydraulic volumetric displacement due to lining wear.

3.3.5.8 Lining Wear Indication. - Each brake shall contain a visable means of indicating the maximum permissible lining wear.

3.3.5.9 Piston "O" Rings. - Operating temperatures of "O" rings shall be 300°F for continuous use. Short term (1 hour) exposure to 450°F shall not result in leakage at maximum operating brake pressure. The "O" rings must last the life of the carbon heat sink without leakage in service.

3.3.5.10 Brake Release. - The brake assembly shall fully release to provide proper running clearance. The released condition of the brake is defined as that condition which exists upon completion of the following sequence:

(a) Apply pressure Pa to inlet port. The value P_a shall be any value from 110 percent of maximum steady state back pressure in accordance with 3.4.2(b) herein to maximum parking pressure of 3.4.2(c) herein.

(b) Reduce applied pressure P_a until 110 percent of maximum steady state back Pressure, in accordance with 3.4.2(b), herein is applied to brake inlet port.

(c) With 110 percent of maximum steady state back pressure in accordance with 3.4.2(b) herein maintained at the brake inlet port; the brake is in the released condition. In the released condition the tangential force shall not exceed 50 lbs. to rotate the wheel.

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3.3.5.11 Bleeder Fitting. - A bleeder fitting will be provided as near to the top of the brake as clearance and service area near the main landing gear will allow. The bleeder fitting in addition to permitting system bleeding, will allow ground installation and removal of bleeder hose standard brake pressure gages without introducing contamination or requiring system rebleeding.

3.3.5.12 Retracters. - Unless shown to be unnecessary, a mechanism shall be provided to fully release torque by retracting the actuator pistons to the designed travel at pressures not less than 130 psi.

3.3.5.13 Brake Stack Clips. - Piston brake stack clips shall last the wear out life of the carbon stack on the aircraft. The clips shall be used to protect the carbon during wheel installation and removal and from friction induced delaminating forces during braking applications.

3.3.5.14 Brake Backup Structure. - The brake back-up structure shall be designed to promote even brake disc pressure and wear radially across the brake disc.

3.3.5.15 Brake Stack Refurbishment. - Supplier is to investigate possible methods and procedures to refurbish carbon heat stacks. The proposed plan is to be submitted with the proposal for evaluation.

3.3.5.16 Torque Characteristics. - The wheel and brake assembly shall be designed to provide the proper balance between the static and dynamic torque requirements; so that adequate static torque is available for carrier usage with a wet brake, while keeping the dynamic torque low enough to prevent over sensitivity for pilot controlled braking (anti-skid off) on normal and low mu runways. With 2700 psig brake pressure, the brake shall not produce a torque greater than its structural torque for any dynamic or static braking condition.

3.3.5.16.1 Static Torque. - The wheel and brake assembly shall be designed to provide the following static torques for the indicated brake pressures. The static torque requirements shall be met at the brake torque just prior to breakaway or the torque just after breakaway (while wheel is creeping) whichever is less. The two engine runup requirement replaces the .55ground coefficient requirement in MIL-W-5013 and shall comply with its demonstration requirements.

REQUIREMENT	HEAT SINK CONDITION	BRAKE PRESSURE	REQUIRED TORQUE
Rolled Deck	Combination of moisture and temperature which results in worst case lining friction coefficient	1800 psig	12600 ft-lb
Two Engine Runup	Dry @ Approx. 300°F	2,700 psig	22,400 ft-lb

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3.3.5.16.2 Brake Pressure-Torque Characteristics. - The wheel and brake assembly shall be designed to provide fast pressure torque response for operation with a low wheel speed deviation skid control system. Special design consideration shall be given to consistent torque gains, elimination of torque sag or grabbiness during all dynamic or static conditions.

3.3.5.17 Stress Analysis - A stress analysis report on all major brake components shall be submitted to NAVAIR for approval.

3.3.5.18 Thermal Analysis - A complete thermal analysis shall be done on all major wheel, brake, and axle components to verify the temperature constraints of this specification are satisfied. The report shall be submitted to NAVAIR for approval.

3.3.5.19 Brake Retention. - The brake assembly shall be designed to be retained on the axle at wheel removal. The brake retention shall be a quick release design.

3.3.6 Identification and Marking.

3.3.6.1 Identification and Markings shall be as noted in paragraph 3.9 of Specification MIL-W-5013.

3.3.6.2 Other Markings. - All other type markings shall be in accordance with MIL-STD-130.

3.3.6.3 Age Control, Packings and Gaskets. - The wheel and brake assembly shall be supplied with suitable markings showing the date of assembly or reassembly of the equipment in quarter-of-year and year (e.g. - 3Q69, representing the third quarter of 1969). Acceptable methods of marking shall be by decal or indelible ink stamping. The age of the oldest packing and/or gasket, at the date of assembly or reassembly of the unit shall not exceed eight (8) quarters (24 months). (Ref. ANA, Bulletin No. 438).

3.3.7 Moisture Entrapment. - The wheel and brake assembly shall be designed to prevent the entrapment of moisture in any position from fully extended to fully retracted. This may be accomplished by effectively sealing enclosed areas against the entrance of water or by providing adequate drainage. Cork seals, dams, and metal end plugs machined to fit shall not be used.

3.3.8 Total Weight. - The total weight of the wheel and brake assembly shall be a minimum consistant with good design and shall not exceed 165 lbs. The seller shall conduct a weight control program.

3.3.9 Reliability Provisions.

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3.3.9.1 Reliability Program. - Reliability shall be considered a design parameter of major importance and shall receive appropriate controls.

3.3.9.2 Quantitative Reliability Requirement - The mean-time-between-failures (MTBF) shall be 20,000 hours as specified in Standard MIL-STD-781.

3.3.9.3 Operational Service Life. - The brake stack and wheel operational service life will be controlled by the design requirements of Section 3, and the actual service environment. There shall be no damage to the equipment, and the equipment shall perform in accordance with the requirements of this specification during or after the equipment has been exposed to any environmental conditions or any combination of the conditions specified herein. The basic brake frame (basic brake housing, torque tube, basic pressure and backup plates, except carbon heat sink portions) service life shall be 5500 landings including 3500 land-based landings and 2000 carrier landings.

3.3.10 Maintainability. - The equipment shall be designed in accordance with the qualitative and quantitative maintainability requirements stated herein. The equipment shall be capable of being maintained at Organizational, Intermediate and Depot levels of maintenance as defined in Volume II, Chapter 2, OPNAVINST 4790.2A.

3.3.10.1 Detail Maintainability Requirements. - The following detail maintainability requirements shall be demonstrated in accordance with the applicable test of 4.3.1.4.

3.3.10.1.1 Preventive Maintenance. - Preventive maintenance shall be kept to a minimum and shall be limited to visual inspection. All preventive maintenance requirements shall be justified, documented and submitted to the procuring activity for approval.

3.3.10.1.2 Corrective Maintenance.

3.3.10.1.2.1 Organizational Level. - Organizational level maintenance generally shall consist of visual inspection, and removal and replacement of faulty wheel and brake assembly.

3.3.10.1.2.2 Intermediate Level. - It is required that all intermediate level repairs of the wheel and brake assembly generally consisting of repair, bench testing, adjustment, removal and replacement of components, etc., be accomplished within 120 minutes mean time and shall not exceed 150 minutes maximum. The mean time to dis-assemble and re-assemble the wheel assembly and brake assembly at the intermediate maintenance level shall be recorded separately with the total time not to exceed 180 minutes.

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3.3.10.1.2.3 Depot Level. - Depot level maintenance requirements shall be kept to an absolute minimum. Any requirements for depot maintenance shall be justified, documented and submitted to the procuring activity for approval.

3.3.10.2 General Maintainability Requirements.

3.3.10.2.1 Task Criteria. - The maintenance actions required by 3.3.10.1 are to be accomplished with a maximum of two maintenance personnel of no higher than intermediate skill (E-4) for at least 95% of the maintenance tasks.

3.3.10.2.2 Adjustments. - Installation adjustments shall not be required.

3.3.10.2.3 Support Equipment. - Requirements for support equipment shall be held to an absolute minimum. In particular, requirements for Peculiar Ground Support Equipment (PGSE) shall be justified, documented, and submitted to the procuring activity for approval prior to first article test.

3.3.11 Safety. - The wheel and brake assembly shall be designed to preclude the incorporation of features which result in critical or catastrophic hazards as classified in MIL-STD-882.

3.3.12 Human Engineering. - The design of the wheel and brake assembly shall conform to the requirements of Standard MIL-STD-1472, paragraph 5.9.

3.3.13 Configuration Approval - Configuration review meetings shall be held periodically before configuration approval is obtained by the seller from the procuring activity.

3.3.14 Environmental.- The equipment shall not suffer damage, deterioration or degradation of performance, beyond the limits of this specification, when subjected to any environment or any natural combination of environments specified herein and in MIL-STD-210. The envdronmental requirements are design conditions and may be verified by similarity and analysis as approved by NAVAIR; no specific additional tests should be required except as noted herein.

3.3.14.1 Temperature. - The equipment shall be capable of meeting the requirements of this specification during and after exposure to the following ambient air temperatures:

(a) Storage: -80°F to + 185°F

(b) Operating: -65°F to 160°F from sea level to 15,000 feet

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(c) Transient - At a stabilized temperature of 160°F the equipment must be capable of sustaining temperature transients where the ambient air and structural skin temperatures in the landing gear compartment will change in five (5) minutes to 330°F and be maintained at this level for an additional 10 minutes and then decrease to 160°F in 5 minutes. Transient cycle life is 72 cycles.

Operation of the brake shall be verified under extreme conditions according to paragraph 4.5.16 of MIL-W-5013.

3.3.14.2 Altitude.- The equipment shall be capable of meeting the requirements of this specification during and after exposure to pressures encountered from sea level to 70,000 feet altitude, with a rate of change of 40,000 ft/min maximum.

3.3.14.3 Humidity. - The equipment, under both operating and non-operating conditions, shall be capable of meeting the requirements of this specification during and after exposure to relative humidities up to 100 percent. This includes conditions wherein condensation takes place in and on the equipment.

3.3.14.4 Salt Atmosphere. - The equipment, under both operating and non-operating conditions, shall be capable of meeting the requirements of this specification during and after exposure to salt-sea atmosphere as encountered in carrier service.

3.3.14.5 Fungus.- The equipment, under both operating and non-operating conditions, shall be capable of meeting the requirements of this specification during and after indefinite exposure to fungus growth as encountered in tropical climates.

3.3.14.6 Sand and Dust. - The equipment, under both operating and non-operating conditions, shall be capable of meeting the requirements of this specification during and after exposure to sand and dust particles as encountered in desert areas.

3.3.15 Vibration and Shock.

3.3.15.1 Transient Loading. - Special attention shall be given to assure that the brake running clearance will not be changed due to "g" forces in any landing gear position. The wheel and brake assembly shall be capable of simultaneously withstanding +190 "g's" limit acceleration in the radial direction (landing) and 6200 radians per second squared in the rotational direction without impairing the function of the wheel and brake assembly. The forcing function for the radial acceleration shall approximate a sine wave having a frequency of 23 Hz. The rotational

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acceleration shall approximate a half sine wave having a time duration of 70 milliseconds. The brake shall perform satisfactorily in the normal aircraft environment during service.

3.3.16 Configuration Management. - Articles furnished in accordance with this specification shall be configured and produced under a system of Configuration Management.

3.4 Performance. - The components of the wheel and brake assembly shall meet the requirements of the sub-paragraphs of 3.4 at the guaranteed weight. Proof of compliance with this requirement consists of final procuring activity approval of the design stress analyses and successful completion of the seller-conducted tests and the procuring activity conducted tests of Section 4.

3.4.1 Wheel Load Spectrum.- The wheel assembly shall be capable of meeting the following:

(a) Rated Static Load - 30,200 lbs.

(b) Design Landing Radial Load - 96,500 lbs. (Test per 4.5.5.1 of Specification MIL-W-5013).

(c) Design Landing Radial Load - 126,000 lbs. (Test per 4.5.5.2 of Specification MIL-W-5013).

(d) Design Cable Radial Load - 160,000 lbs. (Test per 4.5.5.3 of Specification MIL-W-5013).

(e) Combined Radial Side Load

(1) Out board acting side load	Ultimate
(a) Vertical	59,000 lbs.
(b) Side	35,400 lbs.
(2) In board acting side load	Ultimate
(a) Vertical	59,000 lbs.
(b) Side	47,200 lbs.

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3.4.2 Brake Operating Pressure. - The brake operating pressure shall be as follows:

(a) Maximum operating pressure 2700 psi

(b) Maximum steady state back pressure at the brake inlet is 130 psi

(c) Maximum parking pressure is 3000 psi

(d) Minimum parking pressure is 1800 psi

3.4.2.1 Static Torque. - The brake shall have sufficient torque capability to successfully perform the static torque test in accordance with paragraph 4.5.11e of Specification MIL-W-5013 and 3.3.5.16.1 of this specification.

3.5 Workmanship. - Workmanship shall be in accordance with high grade aircraft practice and of quality to assure safety, proper operation, and service life. Workmanship shall be subject to the inspection and approval of the cognizant inspection activity.

4. Quality Assurance Provisions.

4.1 Seller's Quality Program.- Articles furnished in compliance with this specification shall be produced under the control established in the Seller's (See Definitions, Section 6), Quality Program Plan. The plan shall be prepared and submitted to the procuring activity for approval.

4.1.1 Carbon Quality Control. - Seller shall submit with the proposal, a Quality Control Program he will establish to maintain the same quality of carbon material that is used in the qualification test of paragraph 4.3.

4.2 Responsibility for Inspections and Tests - Unless otherwise specified in the contract or purchase order, the seller shall be responsible for the performance of all inspections and tests specified herein. All procedures, tests, data and reports generated, conducted, provided and submitted in accordance with the requirements of this specification shall be consistent with the definitions and ground rules contained in Section 6. The seller shall utilize the facilities of a laboratory acceptable to the procuring activity. The procuring activity reserves the right to repeat any of the inspections and tests set forth in this qualification where such are deemed necessary to assure that supplies and services conform to prescribed requirements. The representatives of the procuring activity and the government inspector shall be advised two weeks in

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advance when tests are to be performed so that representatives may be designated to witness or supervise the tests when so desired. In addition, the procuring activity reserves the right to witness the tests, disassembly and subsequent investigations conducted by the seller on non-conforming articles returned by the procuring activity.

4.2.1 Classification of Tests. - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests	4.3
(b) Acceptance Tests (Quality Conformance)	4.4
(c) Special Tests	4.5

Acceptance of articles under the test provisions specified herein shall not relieve the seller of responsibility to meet the requirements of this specification.

4.3 Preproduction Tests. - Preproduction tests shall be made on equipment completely representative of the equipments to be supplied under contract. Preproduction tests shall be performed in accordance with and meet all requirements of specification MIL-W-5013 except as modified herein.

4.3.1 Scope of Preproduction Tests. - Preproduction tests shall include all tests specified herein and shall be performed in accordance with requirements of this specification. Test values and parameters shall be designated in the design requirements paragraphs of this specification. If during the course of running preproduction tests, corrective action is required, the seller shall re-evaluate the previously approved tests, test plans, and reports to assure their continued validity. The corrective action, extent of required retest, and test plan revisions shall be subject to approval of the procuring activity. Preproduction tests shall consist of all the testing required by this specification.

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4.3.1.1 Acceptance Tests (Quality Conformance). - The acceptance tests shall consist of the tests of 4.4.1 and shall be performed on all preproduction equipment.

4.3.1.2 Design Integrity Test.

4.3.1.2.1 Radial Load Test. - The radial load test shall be performed in accordance with paragraph 4.5.5 of MIL-W-5013 except as modified herein. Bearing cups, cones and rollers shall not be replaced during this test. The use of strain gages and special coatings to determine region of highest stress within the wheel structure is mandatory.

4.3.1.2.1.1 Yield Radial Load. - This test shall be conducted per paragraph 4.5.5.1 of MIL-W-5013 with an applied load of 96,600 lbs. and initial tire pressure of 350 psi ± 10 psi. The test shall be performed with the brake assembly installed and bolts torqued within the normal torque range.

4.3.1.2.1.2 Ultimate Radial Load Test. - The same wheel, with a brake installed, as used for the yield radial load test shall sustain an ultimate radial load of 126,000 lbs., for 10 seconds, without failure. Upon successful completion of this test the brake assembly shall be removed, the radial load reapplied to failure. The load at which failure occurs shall be recorded. Initial tire inflation pressure shall be 350 ± 10 psi.

4.3.1.2.1.3 Design Cable Radial Load Test. - A new wheel, with a brake installed, shall sustain a load of 160,000 lbs. when loaded through a 1 3/8 inch diameter cable or steel bar. Initial tire inflation pressure shall be 350 ± 10 psi. If a cable is used, it should be at least 3 feet long with the ends secured to prevent looseness or unraveling. The load shall be supported for not less than 10 seconds and the resulting permanent set shall not produce loose bearing cups, air leakage or interference in critical running areas. At the completion of the above, the same wheel and brake assembly shall be rolled a minimum of 5000 feet at 30,200 lbs. with the tire inflated to 350 ± 10 psi. It is permissable to compensate tire inflation pressure for flywheel curvature. Distance for each roll cycle shall be 300 feet minimum.

4.3.1.2.2 Combined Radial-Side Load Test. - The combined radial side load test shall be performed in accordance with paragraph 4.5.6 of MIL-W-5013 except as modified herein. The tests shall be conducted with bearing cups, cones and rollers installed. Use of strain gages and special coatings to determine region of highest stress within the wheel is mandatory. Initial tire inflation pressure shall be 350 ± 10 psi.

4.3.1.2.2.1 Yield Combined Radial-Side Load Test. - A new wheel, with a brake installed, shall support three successive load applications of 45,300 lb. radial and 27,200 lb. side (load directed outboard) followed

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by three successive load applications of 45,300 lb. radial and 36,200 lbs. side (load directed inboard) at the 0° position (it shall be the most critical load contact point which will likely include the filler valve hole). Each load application shall be sustained for a minimum of 10 seconds. The successive loadings at the 0° position shall not cause apparent permanent set increments of increasing magnitude. The resulting permanent set shall not produce loose bearing cups, air leakage through the wheel or past the wheel seal, or interfere in any critical clearance areas or render the wheel or brake unsuitable for further service. Initial tire inflation pressure shall be 350 ± 10 psi.

4.3.1.2.2.2 Ultimate Combined Radial-Side Load Test. - The same wheel, with a brake installed, that successfully sustained the yield combined radial-side load test shall be subjected to one application of 59,000 lbs. radial and 47,200 lbs. side (load directed inboard) at the 0° load position. The load shall be sustained for 10 seconds without failure. Initial tire inflation pressure shall be 350 ± 10 psi.

Upon successful completion of the above, the brake shall be removed. With the side load component held constant, the vertical load shall be increased until failure occurs. The failure load shall be recorded.

4.3.1.2.3 Burst Test. - The wheel shall be subjected to the burst test in accordance with paragraph 4.5.7 of Specification MIL-W-5013. The wheel shall withstand a pressure of 1102 psi ± 10 psi or the burst strength of the tire, whichever is lower, provided the burst strength of the tire exceeds 980 psi ± 10 psi.

4.3.1.2.4 Roll Tests. - The wheel shall withstand roll tests performed in accordance with paragraph 4.5.8 of Specification MIL-W-5013 except as modified herein.

A new wheel assembly shall be subjected to thermal conditioning equivalent to the cumulative temperature-time history resulting from brake heat dissipation experienced during the Normal Energy Dynamic Torque Test of 4.3.1.2.8.1. Thermal conditioning may be accomplished by performing dynamic torque testing or by a suitable oven heat soak. Following thermal conditioning, the wheel shall successfully complete the following roll test spectrum, twice in order shown for a total of 3000 miles.

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Condition	Vertical Load	Side Load	Distance	Tire Inflation Pressure
Straight roll	30,200 lbs.	0 lbs.	873 miles	245 psi
Straight roll	30,200 lbs.	0 lbs.	481 miles	350 psi
Reduced Tie Bolt Torque	30,200 lbs.	0 lbs.	25 miles	245 psi
Reduced Tie Bolt Torque	30,200 lbs.	0 lbs.	25 miles	350 psi
Inboard Yaw	40,300 lbs.	-10,100 lbs.	37.5 miles	245 psi
Outboard Yaw	20,100 lbs.	5,000 lbs.	37.5 miles	245 psi
Catapult	50,000 lbs.	0 lbs.	20 miles	350 psi
Catapult	60,000 lbs.	0 lbs.	1 mile	350 psi
			1500 miles	

Notes:

1. Bolt torque during reduced tie bolt portion of test shall be 90% of minimum recommended on the wheel assembly drawing.

2. A maximum of 10 bolt retorques, for whatever reason, is allowed during this 1500 miles.

3. During Catapult Roll, the wheel speed and duration shall be as required for maximum tire life. Roll cycle distance shall be 300 feet minimum.

4. Combined load achieved by yawed rolling.

The wheel must be inspected at the conclusion of the 3000 miles by Zyglo and Eddy Current and found to be free of cracks or other evidence of failure. The wheel will then roll an additional 3000 miles or failure, whichever occurs first. Tie bolt or bearing failure after the first 3000 mile portion of the test has been completed will not be construed as wheel failure.

4.3.1.2.5 Static Test. - The wheel shall be subjected to the test in accordance with paragraph 4.5.10.1 of Specification MIL-W-5013.

4.3.1.2.6 Diffusion Test. - The wheel shall be subjected to the test in accordance with paragraph 4.5.10.2 of Specification MIL-W-5013.

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4.3.1.2.7 Dynamic Pressure Test. - The wheel shall be subjected to the test in accordance with paragraph 4.5.10.3 of Specification MIL-W-5013.

4.3.1.2.8 Dynamic Torque Test. - The wheel and brake assembly shall be subjected to a dynamic torque test in accordance with paragraph 4.5.11 of MIL-W-5013 and as specified herein. The wheel, brake and tire are expandable after an RTO or turnaround test, but the axle temperature shall not exceed 500°F at the torque splines following any braking condition defined herein. The fuse plugs shall not release until a minimum of one minute following the RTO or turnaround test. A maximum of 10 MPH auxiliary cooling air may be used during the dynanometer stops to simulate actual landing conditions, providing it be directed parallel to the rotating plane of the wheel. No additional auxiliary cooling air shall be used until all temperatures have stabilized or are declining except for the 6 minute cool down period of the turn-around condition. A thermal analysis shall be submitted, with the proposal, as initial verification that the temperature constraints noted above are satisfied. Ambient temperature shall be assumed as 75°F.

4.3.1.2.8.1 Normal Energy Dynamic Torque Test. - The normal energy dynamic torque test shall consist of 50 braked stops performed in accordance with Table II, herein. Five Condition A and one Condition B stop shall be conducted after spraying 2 liters of sea water or equivalent over the heat sink with the wheel removed. Static torque pulls at 1000,2000 and 2700 psi shall be conducted prior and subsequent to each of the wet stops and after the dry stops noted by Table II to demonstrate compliance with Paragraph 3.4.2.1 requirement. The tire bead seat temperature shall not exceed 350°F as a result of brake heat developed during any stop performed during the normal energy dynamic test.

4.3.1.2.8.2 Maximum Energy Dynamic Torque Test. - (Rejected Take-Off) Following the 50 normal energy stops, the brake shall be disassembled and inspected. After inspection, the brake shall be reassembled, using the same heat sink components and a Condition D stop conducted.

Following the above RTO, the brake shall again be disassembled and inspected. The brake will again be reassembled with the same heat sink components (drive clips may be replaced if necessary), the brake wetted with two litters of sea water and the brake be subjected to a second Condition D stop for information only.

4.3.1.2.8.3 Turnaround Test. - The following test shall be performed on the Service Spectrum test brake after completion of tests per paragraph 4.3.1.2.9 and the worn brake structural torque test per paragraph 4.3.1.2.13.

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The specific requirements of the Brake Dynamometer tests are: (Dynamometer IE 27,800 lbs.)

(a) A 10,000,000 ft-lb. stop from 104 MPH at a deceleration rate of at least 10 ft/sec^2.

(b) Accelerate to 28.4 MPH, taxi 10,000 ft and stop (.74 X 10^6 Ft-Lb).

(c) Remain stopped X minutes.

(d) Accelerate to 32.5 MPH, taxi 10,000 ft and stop (.98 X 10^6 Ft-LB)

(e) Accelerate to 149.3 MPH and conduct RTO stop of 20,700,000 Ft-LB in 2966 ft (Condition C, Table I)

X = 6 min - time to accelerate dynamometer to 149.3 MPH.

NOTE: During (b) and (d) taxi tests, add 1.5 X 10^6 Ft-Lb by dragging brake during taxi.

4.3.1.2.8.4 Emergency Swept Wing Stop. - Perform an emergency swept wing stop of 21,100,000 ft-lbs with brake application speed of 192 mph. (Test E, Table I). New friction material may be installed if necessary.

4.3.1.2.8.5 Brake Fusing Level Stop. - Following a normal landing at maximum field landing weight, the brakes shall be capable of decelerating the aircraft down to taxi speed within 5000 feet from touchdown and then decelerating to zero within 500 feet. After applying brake pressure for one minute, the brakes shall not fuse or lock with wheels free to roll for continued taxiing. Standard day conditions, no wind, and 0.3 tire coefficient shall apply. Taxi speed is 30 mph.

4.3.1.2.9 Service Wear and Performance Testing. - A wheel brake assembly shall be subjected to a brake life spectrum as defined by Table III. The objective of this test is to demonstrate service wear and performance under wet and dry conditions for carrier and land based operations. Dynamometer inertia equivalent shall correspond to Condition A of Table I except for the carrier deck wet brake taxi series. (Condition D; IE = 34,500 lbs.).

4.3.1.2.10 Static Torque Test - Wet, dry, hot and cold static torque pulls at pressures of 1000, 2000 and 2700 psi shall be conducted as noted by Tables II and III. The test results will be used to verify that the pressure torque requirements of 3.4.2.1 are met.

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4.3.1.2.11 Parking Torque Test. - With the minimum parking pressure in accordance with 3.4.2(d) and 3.3.5.16.1 herein, the brake shall withstand at least 12,600 foot-pounds applied static torque without relative motion between the friction surfaces occurring.

4.3.1.2.12 Taxi and Parking Test. - Within one minute after completing a braked stop performed in accordance with Test Condition A of Table II, herein, the wheel and brake assembly shall be taxied at 20 MPH to 30 MPH for 10,000 feet minimum. Throughout the taxi, pressure not less than 110 percent of the maximum steady state back pressure in accordance with 3.4.2(b) herein, shall be maintained at the brake inlet port. When the taxi is completed, parking pressure in accordance with 3.4.2(c) herein, shall be applied at the brake inlet port and maintained for at least one hour. After at least one hour has elapsed, pressure at the brake inlet port shall be reduced to 110 percent of the maximum steady state back pressure in accordance with 3.4.2(b) herein. The brake shall comply with 3.4.2(c) herein.

4.3.1.2.13 Structural Torque Test. - A new brake shall demonstrate sufficient integrity to sustain a structural torque test performed in accordance with paragraph 4.5.12 of MIL-W-5013, except that the brake shall be pressurized to 4500 psi.

After 750 stops, the worn brake from Service Wear and Performance Testing, paragraph 4.3.1.2.9, shall demonstrate 80 percent of the new brake requirement without failure with the brake pressurized to 4500 psi.

4.3.1.2.14 Endurance. - The endurance test shall be performed in accordance with paragraph 4.5.15 of Specification MIL-W-5013.

4.3.1.2.15 Extreme Temperature Test. - The extreme temperature test shall be performed in accordance with paragraph 4.5.16 of Specification MIL-W-5013. The brake filled with operating fluid shall be subjected to a temperature of 300°F.

4.3.1.2.16 Piston Stops and Proof Pressure Test. - The piston stops and brake housing shall demonstrate their ability to withstand 4500 psi for 5 minutes without the brake discs installed. No deformation or performance degradation shall result.

4.3.1.2.17 Burst Test. - The brake assembly with a piston travel comparable to a "maximum worn" stack shall be subjected to 6000 psi burst test for 5 minutes at room temperature of 75°F.

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4.3.1.2.18 Anti-Skid Compatibility Testing. - Dynamometer testing shall be conducted to determine the compatibility of the F-14 anti-skid system and the carbon brake designed to this specification. As a minimum the following dynamometer stops shall be run. Simulate aircraft hydraulic system. Run tests at high psi/sec application rate.

Brake Application Speed	Brake Stack Condition	Runway Condition	Brake Pressure
120-100 kts.	dry	dry	minimum to activate system
70-60 kts.	wet	dry	maximum
120-60 kts.	wet	wet	maximum
120-60 kts.	dry	wet-dry (if possible)	maximum
120-60 kts.	wet	wet-dry (if possible)	maximum

The following information must be recorded during testing, wheel speed, brake pressure, and valve current. A test plan must be coordinated with and be approved by NAVAIR before start of testing.

4.3.1.2.19 Anti-Spin Braking Test. - A wheel, brake, and tire assembly shall be spun up to 192 mph, then unlanded and immediately apply 2700 psi to the brake to stop the wheel. The onset pressure rise rate shall be 10,000 psi/sec minimum. There shall be no permanent deformation or evidence of structural failure.

4.3.1.2.20 Field Demonstration. - At completion of laboratory testing of the wheel and carbon brake, 5 carbon brakes and 5 wheel assemblies shall be forwarded to the Naval Air Test Center, Patuxent River, Maryland for completion of qualification testing on the F-14 aircraft before final approval. Testing will consist of anti-skid compatability, carrier landing, catapult simulations, service life verification, and brake static holding capability under various brake stack and runway conditions. Supplier shall provide engineering representation during testing as necessary. Proposed Test Plan of Supplier must be coordinated with and approved by NAVAIR before start of testing.

4.3.1.3 Human Factors Engineering Verification Test. - A human factors engineering verification test shall be performed to demonstrate compliance with 3.3.12. The effect of degraded inputs and degraded performance shall be determined.

4.3.1.4 Maintainability Demonstration. - The seller shall conduct a maintainability demonstration in accordance with this specification. The demonstration shall consist of a minimum of three (3) disassemblies and reassemblies of the wheel assembly and brake assembly with the average

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time serving to determine compliance with the requirements of 3.3.10.1.2.2.

4.3.1.5 Reliability Demonstration Test. - The 3000 mile roll test of paragraph 4.3.1.2.4 shall satisfy this requirement.

4.3.2 Preproduction Test Report Appoval - Regardless of any inherent virtue exhibited by the test results themselves, the preproduction test requirements have not been satisfied until the procuring activity has approved the seller's Preproduction Test Report. Any production of equipment prior to such approval is undertaken at the seller's own risk.

4.3.3 Disposition of Test Sample(s) - The equipment tested shall be kept intact by the seller until Preproduction approval has been granted by the procuring activity, after which disposition shall be made in accordance with the contract.

4.4 Acceptance Tests. - Acceptance tests shall be performed to verify that equipment supplied under the contract meet the requirements specified in Section 3, and are equivalent to the preproduction test equipment in all respects, including design, construction, workmanship, test, performance and quality. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of its acceptance in the finished product. All equipments shall have satisfactorily passed the applicable acceptance tests prior to delivery. No deliverable equipment shall have accrued more than 6% of its operating life (including all test and check out time) when received by the procuring activity. If, during testing, 6% of the useful life of a limited life item is exceeded (based on the previously established replacement schedule) those items must be replaced prior to shipment to the procuring activity and a final functional performance check per paragraph 4.4.1.2 must be satisfactorily completed. Evidence of non-compliance with the above shall constitute cause for rejection. For non-complying equipment already accepted, it shall be the obligation of the seller to designate the necessary corrections and incorporate them after approval by the procuring activity.

4.4.1 Scope of Acceptance Tests. - Acceptance testing shall be performed on each production article to determine conformance of the product to the requirements of this specification and shall consist of the following:

4.4.1.1 Examination of Product. - The wheel and brake assembly shall be examined to determine conformance with Figure 2 and 3 and all requirements of this specification for which there are no specific tests.

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4.4.1.2 Functional Tests. - The following functional tests on the wheel and brake assembly shall be conducted in accordance with paragraph 4.5.1 of Specification MIL-W-5013 unless modified herein.

4.4.1.2.1 Brake and Wheel Assembly. - The brake and wheel assembly shall be tested as follows:

4.4.1.2.1.1 Brake. - The brake shall meet the following functional tests.

(a) Actuation 25 times.

(b) Free release, including clearance measurement

(c) Check that adjusters are all stroking evenly.

4.4.1.2.1.2 Static Pressure. - The static pressure shall be in accordance with paragraph 4.5.14 of Specification MIL-W-5013.

4.4.1.2.1.3 Wheel. - The wheel shall be tested to the following requirements:

(a) Radial and Lateral Runout. - In accordance with paragraph 4.5.4 of Specification MIL-W-5013.

(b) Pressure Retention. - In accordance with paragraph 4.5.3 of Specification MIL-W-5013.

4.4.2 Acceptance Tests. - Acceptance tests, consisting of (a) individual tests and (b) sampling plans and tests, shall be the responsibility of the Seller.

4.4.2.1 Individual Tests. - Prior to delivery of all equipment unless otherwise directed by NAVAIR, Seller shall subject the equipment submitted for acceptance under this contract to the inspections and tests performed in accordance with Seller prepared test procedures as approved by procuring activity.A test shall be included to test each individual self adjuster to Seller's established quality control push/pull-through-force limits to insure trouble-free operational service usage.

4.4.2.2 Sampling Plans and Tests. - Prior to equipment delivery, the Seller shall perform sampling tests as specified in MIL-W-5013, paragraph 4.4.2.2 in accordance with a Seller procedure approved by procuring activity. The sampling plan shall include the primary configuration carbon heat sink

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material consistency tests to assure uniformity in friction, wear, oxidation, density, strength, flexibility, etc., compared to the material initially qualified. Representative samples from each manufacturing material lot or batch shall be submitted to tests similar to tests performed on the qualification material samples.

4.4.3 Verification by Analysis, Inspection and Demonstration. All requirements not verified by testing shall be the Seller's responsibility to verify by analysis, inspection or demonstration.

4.5 Special Tests.- Changes in design and material subsequent to completion of the preproduction tests may require additional or revised testing of the equipments involved. Such tests as are directed by the procuring activity for the purpose of evaluating the changes on the performance and quality of the revised equipments are classified as Special Tests.

4.5.1 Scope of Special Tests. - Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the seller shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.6 Equipment Failure - Should a failure occur during any of the Acceptance or Special Tests specified herein, the following action shall be taken:

(a) Immediately notify the procuring activity's representative.

(b) Prepare a malfunction report, noting suspected cause, and submit to procuring activity.

(c) Determine the cause of failure.

(d) Determine if the failure is a recurring manufacturing problem or design deficiency.

(e) Submit analysis to the procuring activity.

(f) Submit to the procuring activity for approval the proposed corrective action intended to reduce the possibility of the same failure(s) recurring.

4.6.1 Corrective Validation.- When a failure occurs during the Acceptance Tests, the schedule shall include a test to check all equipment for the noted non-conformance until it has been determined that the defect has been satisfactorily corrected.

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4.7 Qualification by Similarity.- Wherever adequate factual data exists, demonstrating similarity of design and requirements, this data shall be utilized to minimize the total test program.

4.8 Test Procedures. - Test procedures shall be in accordance with the applicable requirements of Specification MIL-T-18303. The procedures used for conducting Preproduction Tests and Acceptance Tests shall be prepared by the seller and submitted to the procuring activity for approval.

4.8.1 Environmental Test Procedures. - Prior to, during and after each environmental test, readings shall be taken to determine that the operational limits of this specification are met with the exception of salt spray, and fungus for which readings shall be taken only at the start and finish of each test.

4.8.2 Extreme Temperature Clearance Analysis. - The preproduction test procedure shall include a dimensional analysis proving that adequate clearance is available for all parts having relative motion at both temperature extremes under the most adverse dimensional combinations. This analysis shall be prepared so as to satisfy the intent of paragraphs 4.3.1 and 4.3.2 of Specification MIL-H-8775.

4.9 Rejection and Retest. - Equipment which has been rejected may be reworked, or have parts replaced to correct the defects, and resubmitted for acceptance. Before resubmittal, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the procuring activity for approval. After corrections have been made, all tests deemed necessary by the procuring activity shall be repeated.

4.9.1 Defects in Items Already Accepted. - When investigation of a test failure indicates that like defects exist or could exist in items already accepted, the seller shall so advise the procuring activity, designate the necessary corrective action(s) and incorporate them after approval by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 General. - Each wheel and brake assembly shall be preserved, packaged, packed for the level of shipment specified in the contract or purchase order in accordance with Standard MIL-STD-794.

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6. NOTES

6.1 Intended Use. - Wheel and brake assemblies covered by this specification are intended for use on the airplane main landing gear.

6.2 Precedence of Documents. - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) Drawing and This Specification. - This specification shall have precedence over all applicable subsidiary specifications and the specification drawing. When a conflict arises between this specification and the drawing, it shall be brought to the attention of the procuring activity for resolution. Any deviation from this specification or from subsidiary specifications where applicable shall be specifically approved in writing by the procuring activity.

6.3 Performance Objectives. - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified herein can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Definitions. - The following terms are defined for the purpose of this specification.

(a) Procuring Activity - The procuring activity as referenced in this specification shall mean the Naval Air Systems Command.

(b) Seller - The seller is the agency responsible to Naval Air Systems Command for supplying the equipment described in this specification.

(c) Failure - A failure is the inability of the equipment to perform its required function within the specified conditions for a specified duration.

(d) RTO - Rejected Take-Off

(e) Miles - Statute miles

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(f) Operate, Operating, Operation. - When used with no obvious contextual modification, these terms shall be read as if followed by the phrase "in accordance with all requirements of this specification."

6.5 Data Requirements. - The selected data requirements in support of this specification will be reflected in a DD Form 1423 CDRL attached to the request for proposal, invitation for bid, or the contractual document. Unless otherwise specified by the procuring activity, the following data as described in the applicable DD Form 1664 are required.

(a) 3.3.4.11 & 4.8.2 - Report on Radial Clearance between wheel and brake under all loading and temperature conditions. DID Number DI-A-5009A.

(b) 3.3.4.12 - Wheel Stress Analysis Report. DID Number DI-A-5009A.

(c) 3.3.5.17 - Major brake components Stress Analysis Report. DID Number DI-A-5009A.

(d) 3.3.5.18 - Wheel, Brake, and Axle Thermal Analysis Report. DID Number DI-A-5009A

(e) 4.1.1 - Carbon Quality Control Test Plan. DID Number DI-T-5204.

(f) 4.3.1.2.18 - Anti-Skid Compatibility Test Plan. DID Number DI-T-5204.

(g) 4.3.1.2.20 - Field Demonstration Test Plan. DID Number DI-T-5204.

(h) 4.3.2 - First Article Test Report. DID Number UDI-T-21349.

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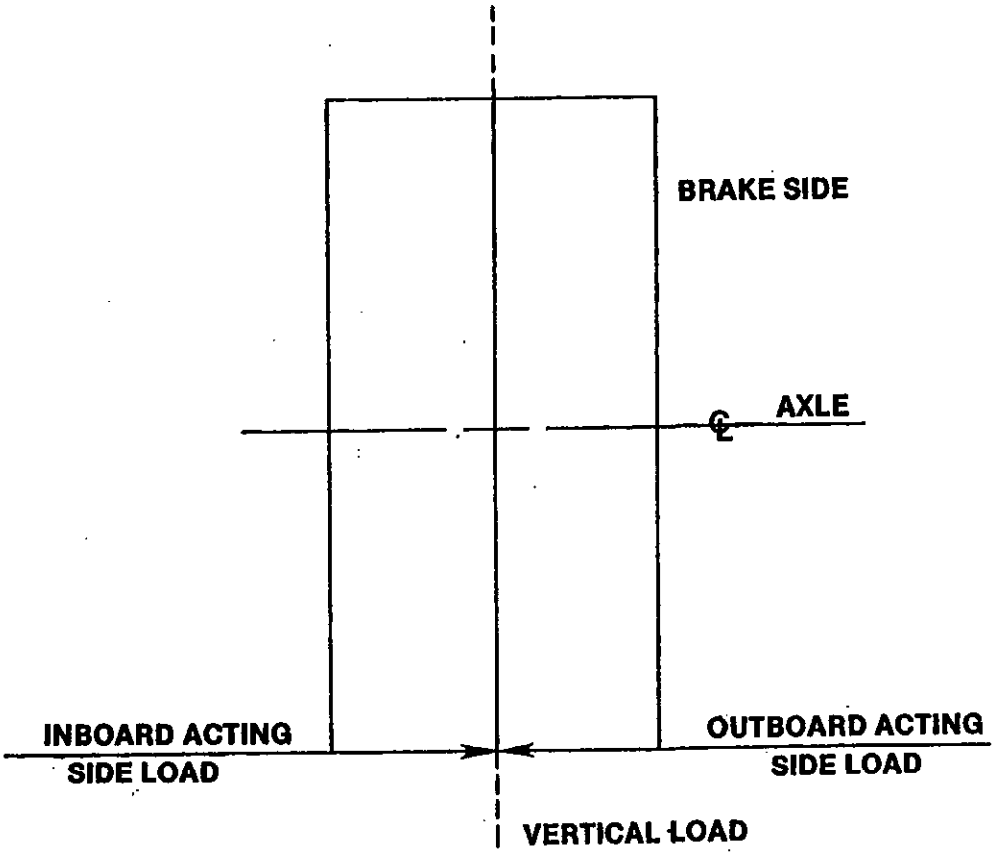


FIGURE 1

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TABLE I

TEST CONDITION	A NORMAL	B OVERWEIGHT	C RTO HOT	D RTO COLD	E EMERG
KINETIC ENERGY FT-LBS	10,100,000	13,500,000	20,700,000	25,500,000	21,100,000
DYNAMOMETER I.E. (LBS)	19,850	23,000	31,000	34,500	17,000
WHEEL LOAD (LBS)	19,900	22,600	26,500	30,100	20,500
TIRE LOADED RAD (INCHES)	16.5	16.2	15.9	15.8	16.2
BRAKE APPLICATION SPEED (MPH)	123.0	132.5	141.0	149.0	192.0
STOP DISTANCE (FEET)	1565	1817	2966	2587	4497
STOP TIME (SECONDS) REF.	17.8	19.3	25.9	21.9	35.2
AIRCRAFT WEIGHT, (Pounds)	49,100	57,100	65,400	71,900	46,000

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TABLE I (Continued)

NOTES:

1. Radial wheel loads listed are to be applied when testing is performed with a loading mechanism which controls radial wheel load independent of tire rolling radius and brake drag force produced. In this case, values for tire loaded radius listed are not applicable.

2. Values for tire loaded radius listed are to be maintained when testing is performed with a loading mechanism which controls tire loaded radius (i.e., locked loaded radius method). If a loaded radius other than minimum listed in Table I is required to prevent the tire from skidding, the stop distance shall be decreased by the ratio of the radius used to the radius in Table I.

3. Value for stopping distance listed is nominal. Actual stopping distance achieved for an individual stop performed in accordance with Test Condition A may exceed nominal value shown by not more than 10 percent provided average of stopping distances achieved for 50 consecutive stops is not greater than nominal value.

4. Values for stop time are presented for reference only. All tests must be performed within required stopping distance regardless of elapsed stop time.

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TABLE II

Stop Number	Dynamic Test Condition (See Table I)	Heat Sink Condition	
1-4	A	DRY	
5	A	DRY	NOTE 1
6-7	A	DRY	
8	B	DRY	
9	A	DRY	
10	A	WET	NOTE 1
11-17	A	DRY	
18	B	DRY	
19	A	DRY	
20	A	WET	NOTE 1
21-24	A	DRY	
25	A	DRY	NOTE 1
26-27	A	DRY	
28	B	WET	
29	A	DRY	
30	A	WET	NOTE 1
31-37	A	DRY	
38	B	DRY	
39	A	DRY	
40	A	WET	NOTE 1
41-44	A	DRY	
45	A	DRY	
46-47	A	DRY	
48	B	DRY	
49	A	DRY	NOTE 1
50	A	WET	NOTE 1
51	D	DRY	
52	D	WET	

NOTE 1: Each stop shall be preceded and followed by static torque pulls at 1000, 2000 and 2700 psi.

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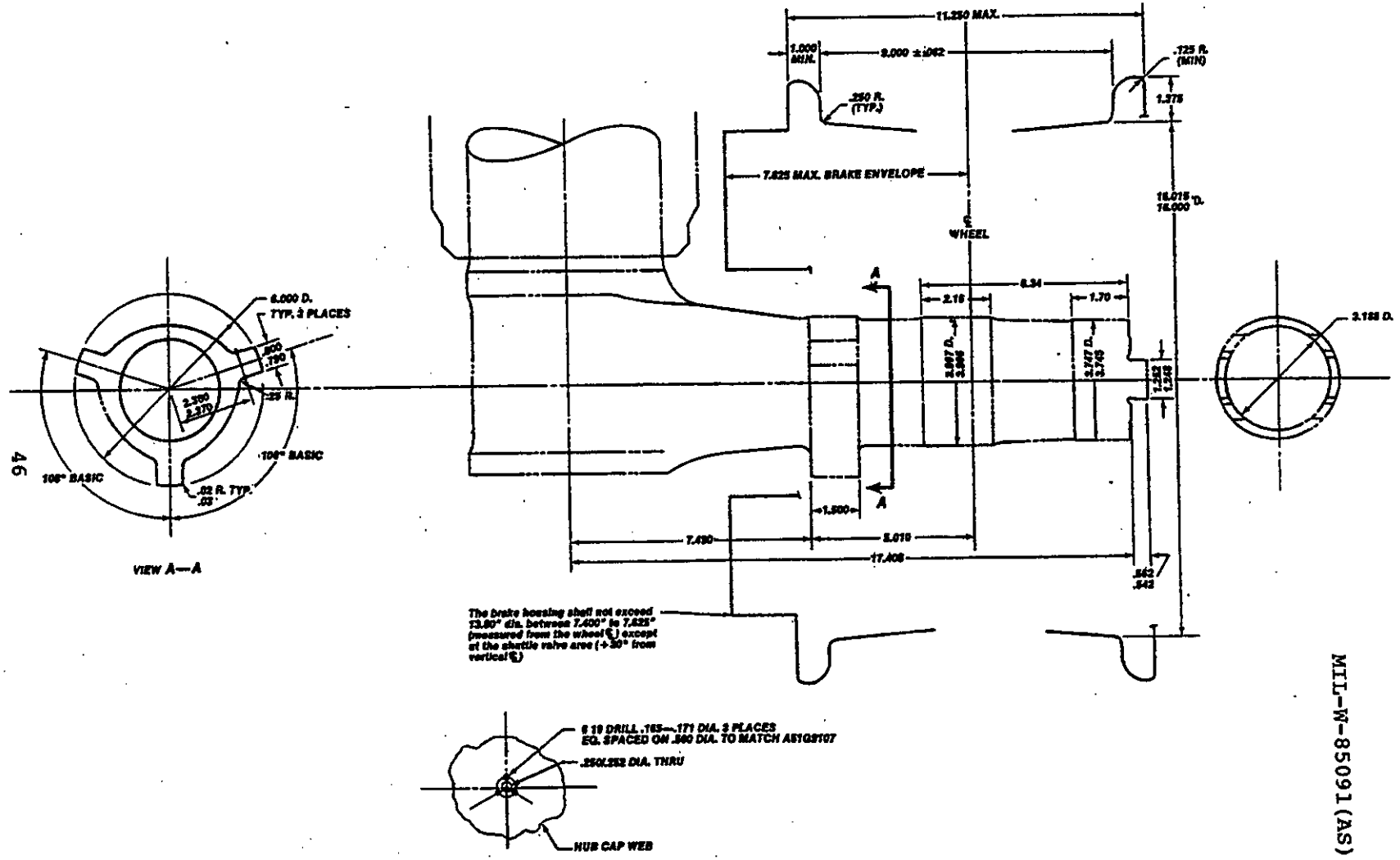
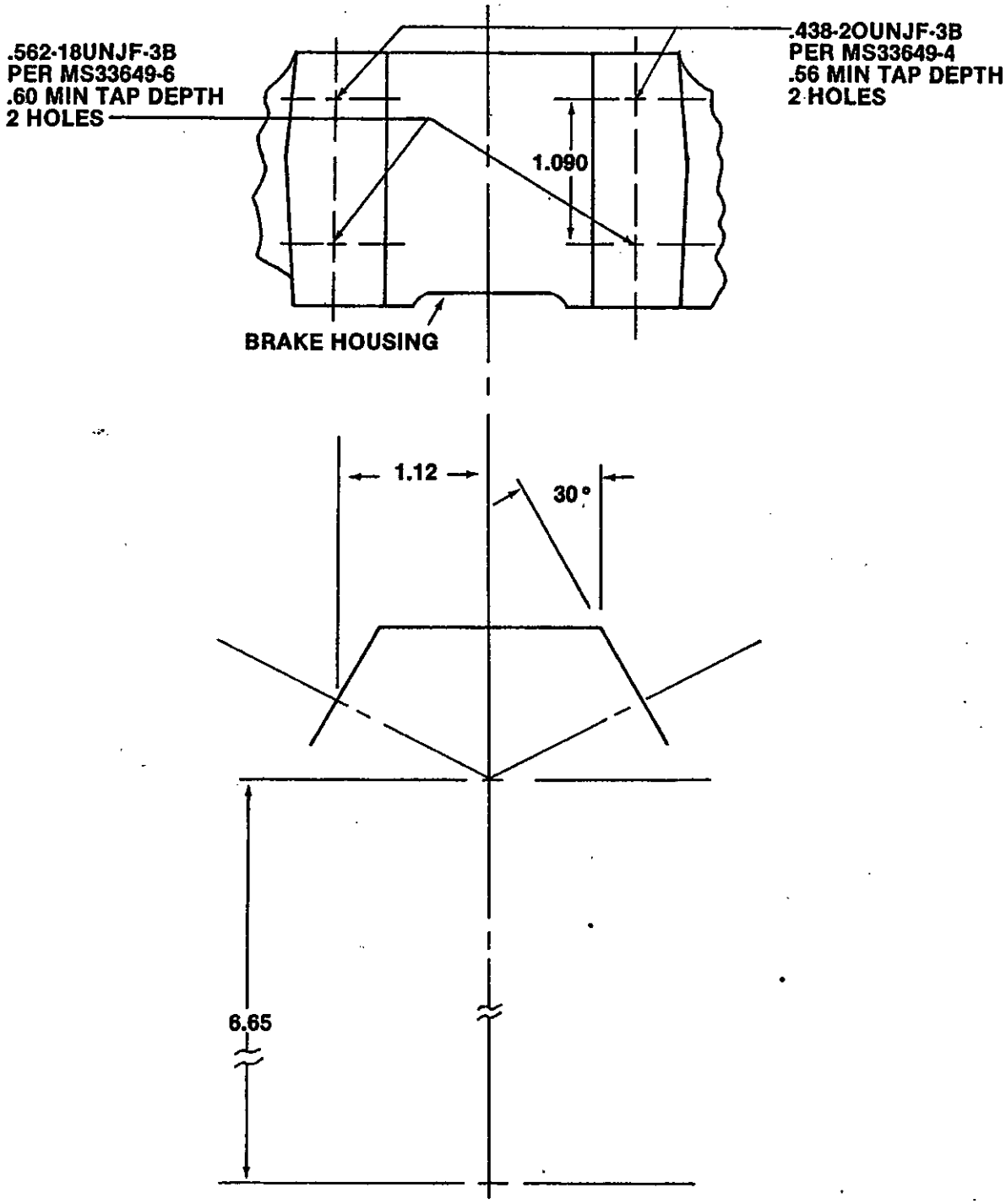


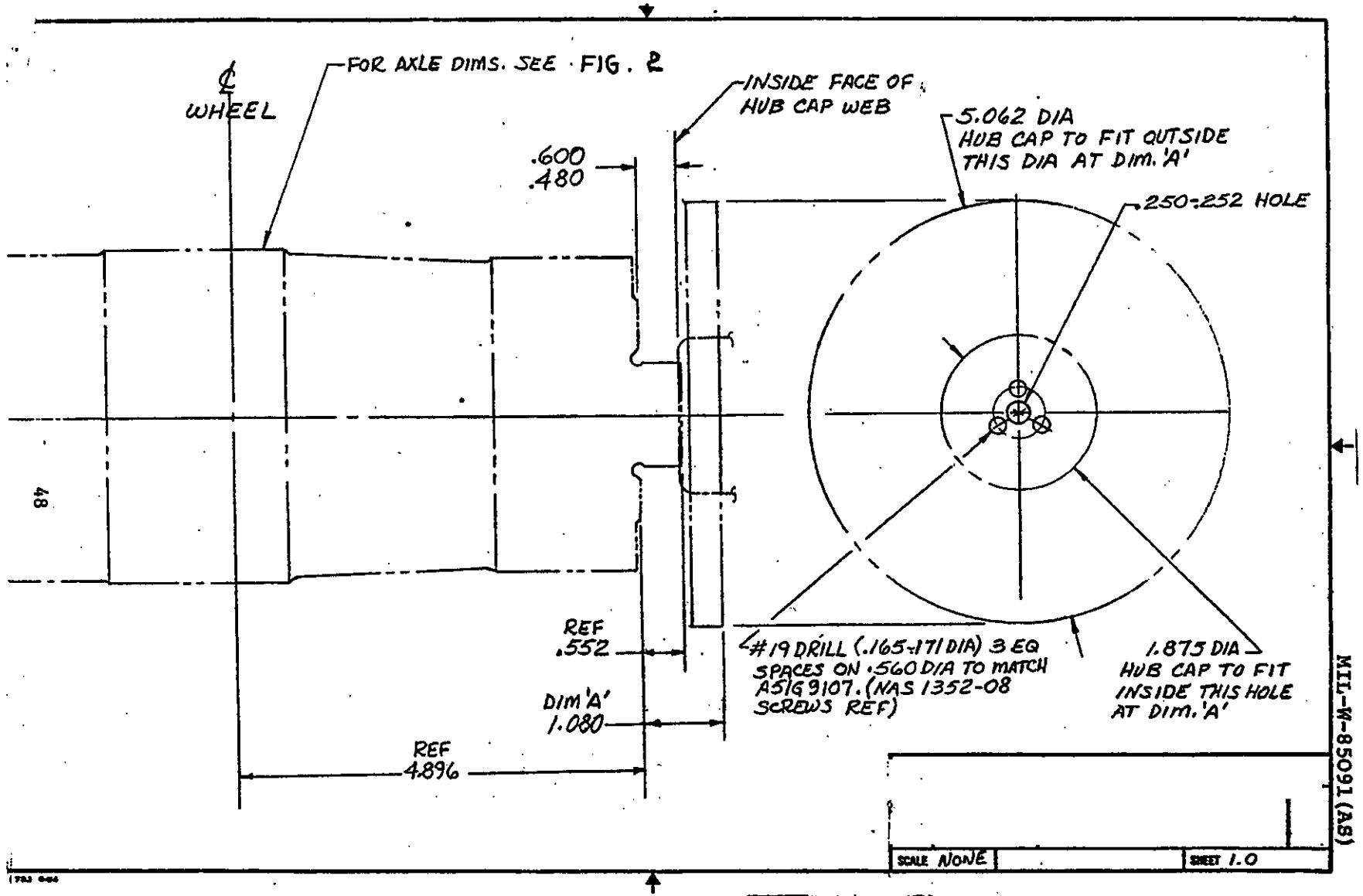
FIGURE 2

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BRAKE INLET LOCATIONS
FIGURE 2 2 OF 2

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HUB CAP INTERFACE DIMENSIONS
FIGURE 1

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TABLE III
SERVICE WEAR & PERFORMANCE SPECTRUM

Brake Condition		1	2	3	4
Stop Description		Service Ldg.	Lt. weight Service Ldg.	Normal Ldg.	Aborted Mission Ldg.
K.E. (primary stop) (FT-LB X 10^6)		6.9	5.6	10.1	13.5
Decel. (fps^2)		5	3	10	10

Perform a total of 750 landing/taxi cycles at each condition as indicated below. All landings to be preceded and followed by a two mile taxi drag at .75 X 10^6 ft-lb/mile and a 30K + -0 taxi stop (4.58 X 10^6 ft-lbs total.) Dynamometer I.E. = 19,850. Initial heat sink temperature prior to primary stop = 150°F. 5 mph cooling air permitted parallel to tire plane.

		1	2	3	4
Start Test	a	75 1 5	-	5	-
Perform from	b	75 2	-	9 2	1
left to right	c	110 3	50 6	7 3	1
and top to	d	40 4	-	3 4 5	1 2
bottom	e	40	50	10	1
	f	110 1	50	17 2	1 2
	g	34 2 5	50 6	9	1
		484	200	60	6

See next page for notes.

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NOTES

1 Prior to this test sequence, conduct twenty 5kt-0 taxi stops with the new heat sink thoroughly sprayed with 4 liters of sea water (or equivalent) prior to each stop. Remove the wheel each time for thorough wetting with brake pressure off. Alternately conduct 5 fps^2 and 3 fps^2 stops with flywheel I.E. = 34,500 corresponding to MTOGW. Record brake pressures and decelerations for each stop. Brake pressure shall not exceed 2700 psi per Para. 3.4.2(a). At the conclusion of twenty taxi stops, conduct wet static torque tests at 1000, 2000, 2700 psi. Rewet prior to each test point.

2 For the 1st stop only, wet the brake with water prior to the landing stop (no taxi after wetting.) Wetting procedure per 1 . Conduct primary stop as soon after wetting as possible. Brake temperature 100°F prior to wetting.

3 For the 1st stop only, wet the brake prior to taxi-out per Note 2.

4 For the 1st stop only, remove the wheel and pour 1/2 liter of MIL-H-5606 over the top of all discs prior to taxi-out.

5 Conduct hot static torques after the last stop at 1000, 2000, 2700 psi. Brake heat sink temperature range of 400 to 500°F.

6 Conduct cold static torques after the last stop at 1000, 2000, 2700 psi and ambient temperature.

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Preparing activity:

NAVY (AS)

Project No. 1630-N302

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